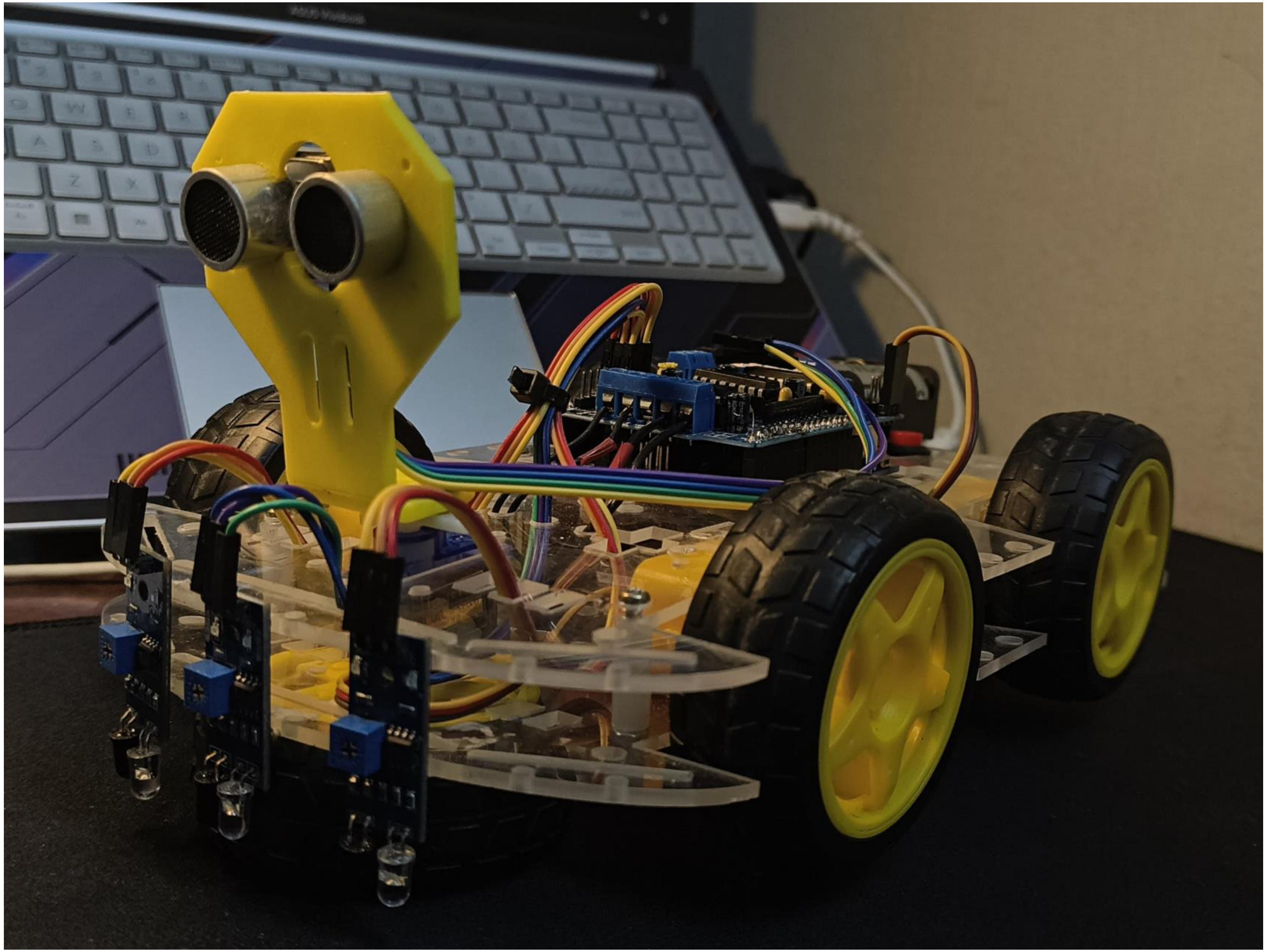
Autonomous Action Oriented Line Follower and Object Detector



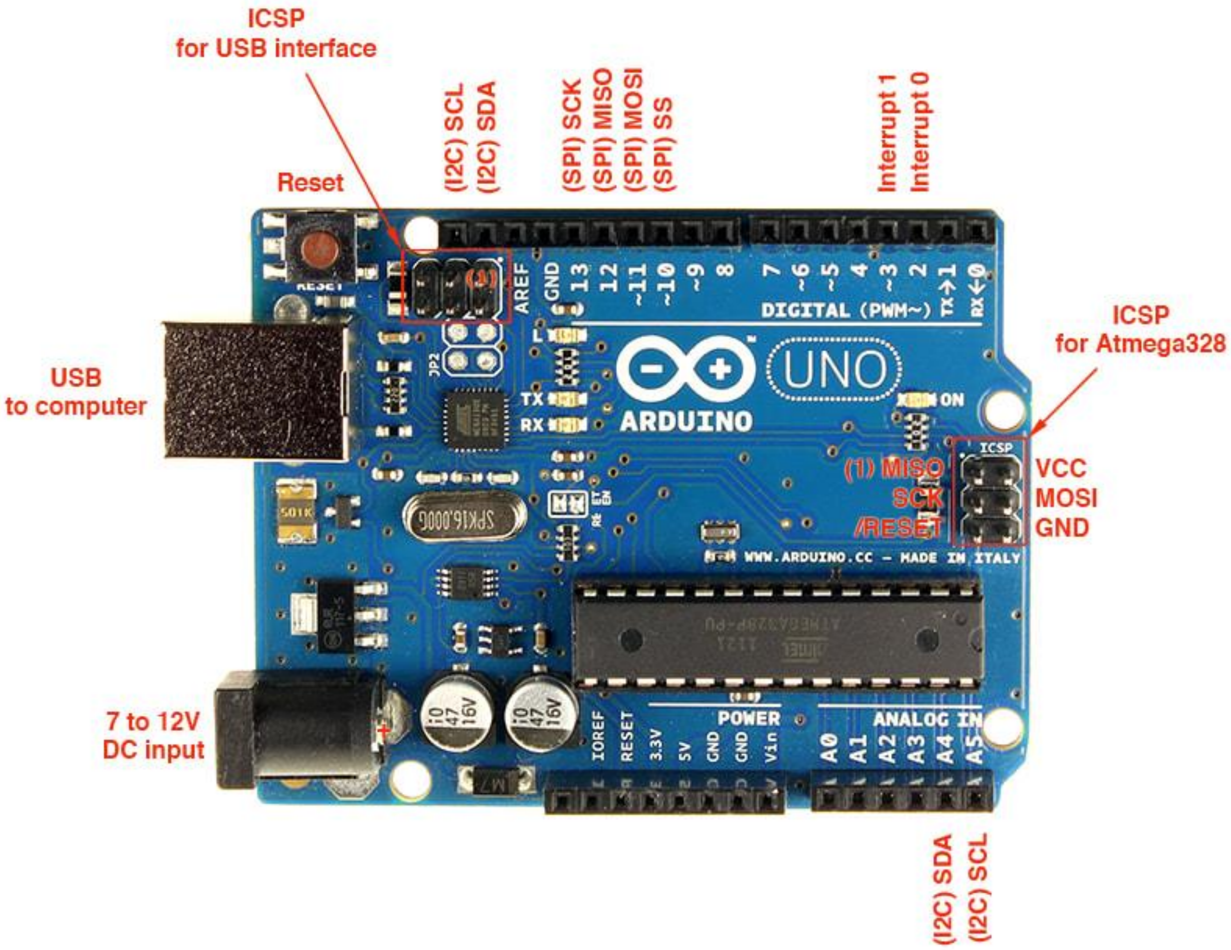
Physics Principles involved:

- ***Centripetal Force*** *– acts during turns to maintain curved motion.*
- ***Friction*** *– enables wheel traction and controlled motion.*
- ***Newton's Laws of Motion*** *– govern the robot's movement and response to forces.*
- ***Momentum*** *– explains continued motion and stopping behavior.*
- ***Torque and Rotational Motion*** *– motor torque converts to wheel rotation and forward motion*
- ***Work, Energy, and Power*** *– relate to motor effort and energy use.*

Applications of the Bot:

- ***Logistics and Supply Chains*** *– can follow preset paths for delivering items between stations.*
- ***Service Robots*** *– delivers food, medicines, or documents in hospitals, offices, or restaurants.*
- ***Surveillance and Inspection*** *– used for monitoring restricted or hazardous areas autonomously.*
- ***Search and Rescue*** *– navigates predefined routes while detecting and avoiding obstacles.*
- ***Agriculture*** *– can be adapted for crop monitoring or automated line-following sprayers.*
- ***Home Automation*** *– as a base for cleaning or delivery robots within households.*

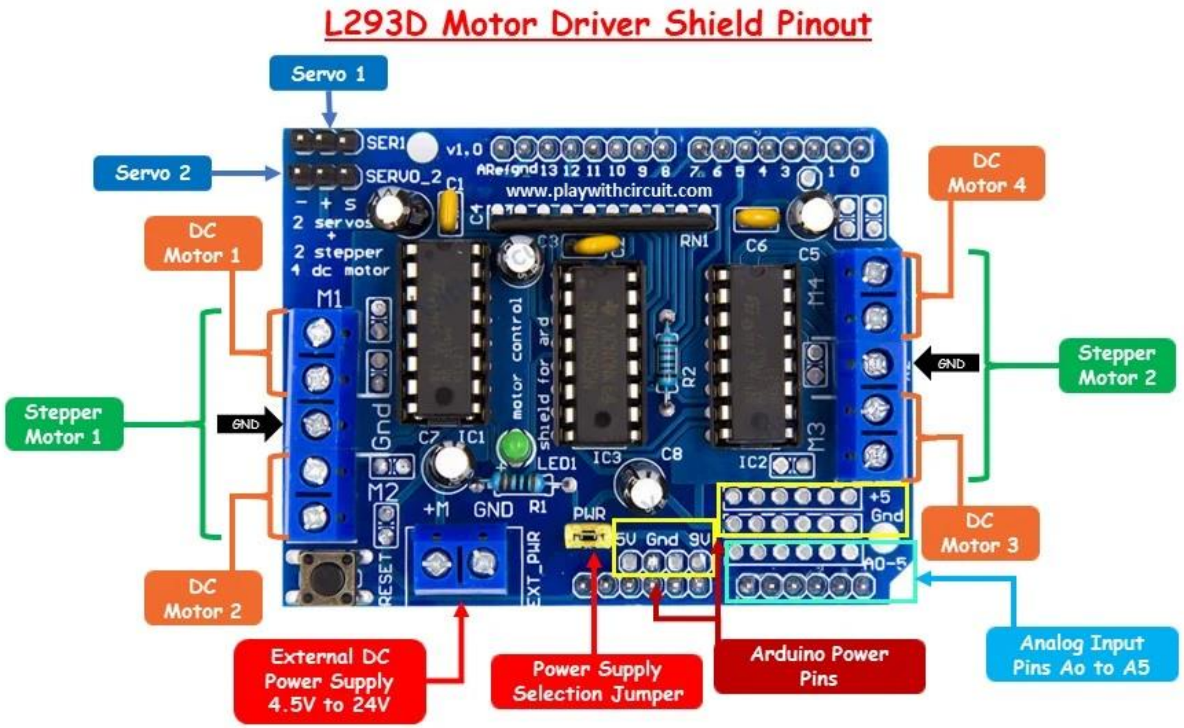
Arduino UNO R3 Micro Controller



The Arduino UNO is the best board to get started with electronics and coding. If this is your first experience tinkering with the platform, the UNO is the most robust board you can start playing with. The UNO is the most used and documented board of the whole Arduino family.

Arduino UNO is a microcontroller board based on the ***ATmega328P.*** *It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. .*

HW-130 Motor Driver Shield



- *HW-130 Motor Driver Shield, L293D Motor Shield (most common due to the main IC). To safely control the **speed** and **direction** of DC motors using a low-power microcontroller (like an Arduino).*
- *The H-Bridge configuration is essential for controlling motor direction.*
- *Controls up to **4 Bi-directional DC Motors** (with 8-bit speed control, using PWM).*
- *OR up to **2 Stepper Motors** (unipolar or bipolar).*
- *AND **2 Servo Motors** (separate connections).*

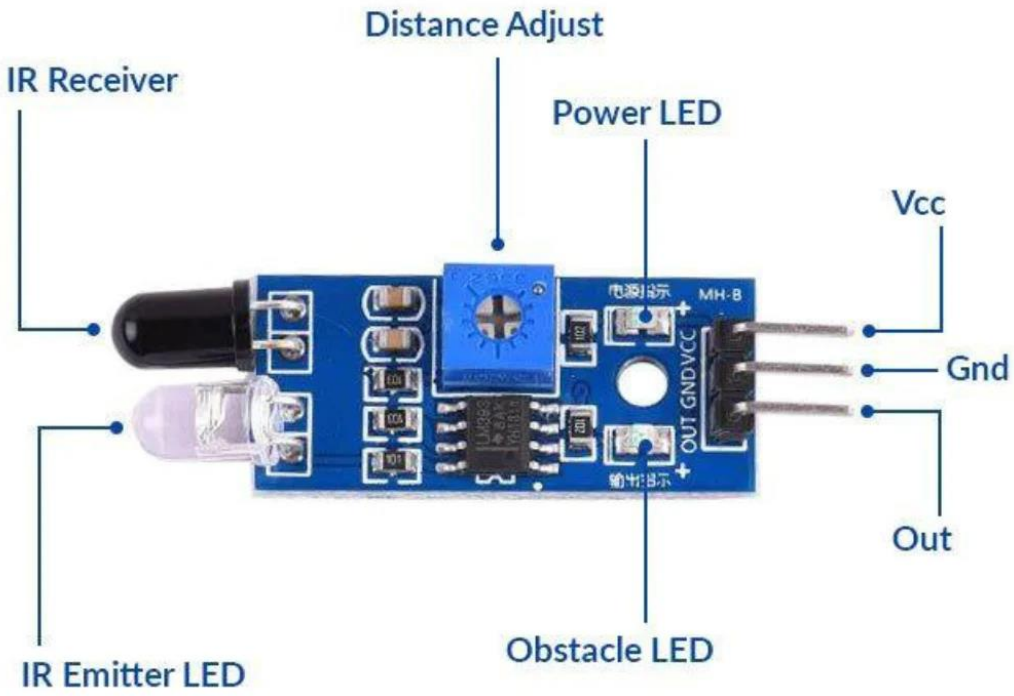
Dual Shaft DC Gear Motor

Locomotion (Movement) : *The main function of the DC gear motor is to* ***drive the robot's wheels****, providing the force and speed needed for movement.*

Torque and Speed: *The integrated* ***gearbox*** *in a gear motor reduces the motor's speed while significantly* ***increasing its torque****.*

In summary, the DC gear motor provides the ***power*** *to move the bot along the line and around obstacles, while the dual shaft design enables the* ***precision*** *and* ***feedback*** *necessary for accurate line following and path adjustments after object detection.*

Infrared Sensor (IR)

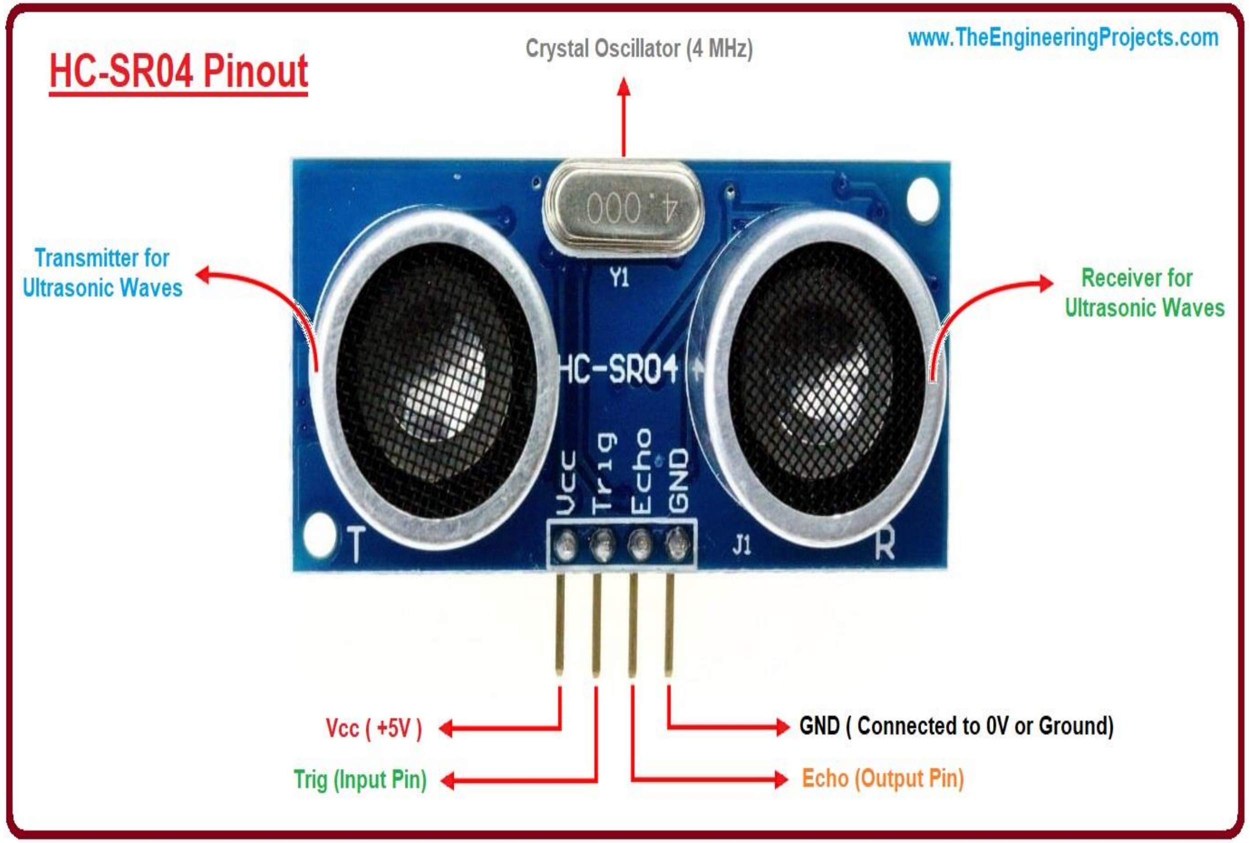


An Infrared (IR) sensor in a line follower and object detection bot acts as the robot's eyes, detecting light reflected from surfaces to provide crucial navigation data.

White surface (or light background): *Reflects most of the IR light back to the receiver, resulting in a* ***LOW*** *or logic '0' signal (or a high analog value, depending on the sensor type).*

Black line (or dark path): Absorbs *most of the IR light, meaning very little is reflected back to the receiver, resulting in a* ***HIGH*** *or logic '1' signal (or a low analog value).*

Ultrasonic Sensor (UR)



Distance Measurement: *Emits ultrasonic sound pulses and calculates the exact distance (in cm) to any object in front of it based on the echo's return time.*

Obstacle Avoidance: *If the measured distance is less than a programmed safety threshold (e.g., 20 cm), the robot pauses its line-following routine and initiates an obstacle avoidance sequence (stop, reverse, turn, and scan for a clear path).*